LiDAR Basic Obstacle Detection Report (Detailed)

1. Introduction

LiDAR is a sensor that measures distances using laser beams. It scans the environment and provides distance readings at different angles, allowing us to detect objects around the sensor.

2. Objective

The goal is to detect obstacles using LiDAR by checking if any object is closer than a specific threshold (50 cm).

3. Concept Explanation

LiDAR gives two main values: Angle (direction) and Distance (how far the object is). If the distance is less than 50 cm, we consider that an obstacle exists.

4. Python Code Implementation

```
# Simple LiDAR Obstacle Detection

for scan in lidar_data:
    angle = scan.angle
    distance = scan.distance

    print(f"Angle: {angle}°, Distance: {distance} cm")

    if distance < 50:
        print("Obstacle detected")
    else:
        print("Clear")
```

5. Experiment

We placed objects like a hand, book, and box in front of the LiDAR sensor and observed how the readings changed depending on the distance.

6. Sample Data

Angle (°)	Distance (cm)	Result
0	120	Clear
45	40	Obstacle
90	30	Obstacle

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180	200	Clear

7. Observations

The system successfully detected nearby objects. However, readings sometimes fluctuated slightly due to sensor noise.

8. Challenges

Noise in readings, difficulty detecting small objects, and environmental effects like reflections.

9. Conclusion

The system worked correctly using a simple threshold. LiDAR is effective but can be improved with filtering techniques.

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